

Web Application & Interface Design Presentation



Overview

Websites are the front door to any business, and especially critical for a restaurant. 90% of customers that visit a restaurant have already researched it online prior to dining there, more than any other type of business. With so much interest and competition, it's extremely important to make the right first impression. This is why the elegant, fine-dining establishment, Café Fausse, has reached out to your team for a brand-new website. They know you can deliver a product that is in line with their own high standards. It looks like you will need to pull out all the stops to impress them!

To do this, you will be developing a complete web application to meet specific software requirements. You are not expected to have carried out significant testing (testing approaches are to be covered in a later concentration), but you should have a complete development version that at least addresses all software requirements.

The web application can be completed **individually** or as part of a **group** of up to no more than 3 people. This deliverable includes a recorded presentation in which the individual student or the group must demonstrate the functionality of their web application.

Learning Outcomes

When completed successfully, this project will enable you to:

- Develop working websites using React and JSX, and Flask
- Translate customer requirements into a working web application
- Carry out high quality UI and UX design
- Develop backend logic and a backend database using Flask and PostgreSQL
- Implement client-side forms with React and form handling with Flask
- Utilize advanced AI tooling for rapid and high quality software development

Project Description

You have been provided both with a copy of the:

- [software requirements specification](#) for this project, and
- a [collection of images](#) for you to use.

You are highly encouraged to utilize leading AI code generation tooling to assist in rapidly producing your Web application, being sure to describe in broad terms how you made use of them. Here are some examples of [very useful AI tools](#) you may wish to consider. You will be graded on the quality and functionality of the application and how well it meets the requirements - no given proportion of the code is required to be hand coded.

AI-assisted software development tooling options include:

- **Cursor**, an AI-based IDE, including agentic capabilities (<https://cursor.com/>) - a good starting point for local development
- **Codex** (<https://chatgpt.com/>) - if your git and GitHub skills are already up to speed
- **Claude Code** (<https://code.claude.com/docs/en/overview>) - if your command line interface, git and GitHub skills are already up to speed
- Another tool of your choice
- During the MS AI Engineering program you will make use of all of the tools listed above - so take this opportunity to go as far as you are comfortable during this project

Hint: Start by loading the software requirements specification document into your AI tool or model of choice, optionally in planning mode, and then enhance, test and modify your code using your selected AI-assisted software development tooling.

You have been tasked with crafting a website from scratch using your knowledge of **HTML**, **CSS**, and **JavaScript**. Restaurants are dependent upon intuitive interfaces and impressive visuals. This is where **React** will come into play. React is a flexible and efficient JavaScript library used to build user interfaces. It accomplishes this by combining small code elements, or “components”, into much larger designs. With an easy-to-learn approach and a wide array of resources and support, React can provide the ability to scale and reuse components across much larger projects. It’s a handy tool to have in your kit. To get started, visit the React tutorial page [here](#).

Work through the tutorials to get familiar with the React library and how to integrate it. Once you are comfortable, your team can begin planning the restaurant site.

First, consider the fundamental functions the customer wants to see on the site:

- Contact information: Address, phone number, hours
- Their menu, broken up by categories
- An “About Us” page highlighting the owners

- An email sign-up for a newsletter
- A photo gallery page, and judicious use of photos throughout the site
- Something showcasing their awards and positive reviews
- An ability to make a table reservation via the Web site

You will also want to provide a constant thematic look and feel by applying a consistent CSS for the entire site. You can either build this directly or utilize CSS-in-JS to code through the React scripts. Your CSS implementation should take either a Flexbox, or Grid approach to design. For more information, check [here](#).

In addition to needing a new web site providing a great web presence, the owner of Café Fausse is tired of fielding so many calls for available tables. What they need is a web-based system that provides the capability to accept reservations for customers. This system should be able to select specific timeslots, input the number of guests, assign a name to the reservation, require an email address and an optional phone number and store this information in the system. It should also check to ensure the timeslot is not fully occupied.

To address the reservation system requested by the owner of Café Fausse, you will need to build both a front-end interface using forms, and a back-end SQL database that stores the reservations. This should be accomplished with a combination of a React front-end form, and using Flask/Python and PostgreSQL for back-end logic and database respectively. Your PostgreSQL database should have, at a minimum, a table for **Reservations** and **Customers**. **Customers** should have a “Customer ID”, “*Customer Name*”, “Customer Email”, “*Phone Number*” and “Newsletter Signup”. **Reservations** should have a “Reservation ID”, “Customer ID”, “*Time Slot*” (date and time), and “*Table Number*”. The back-end logic should take the customer’s information, add it to the customer table, then assign a random available table for the chosen time slot (assume there are 30 tables total). Once completed, it should confirm to the user a successful reservation. If all seats are taken for that time slot, the system should send back a message that the customer needs to pick another time.

In total, your website should have a minimum of five pages:

- Main (index) page
- Menu
- Reservations
- About us
- Gallery

The additional elements listed above may be present on one of these pages or others. The email sign-up should utilize **form** elements to provide this functionality and include basic verification on the input fields, and also store sign-ups in a backend database. Once you are satisfied with the look and feel, test your site to ensure all links work and

the behavior is intended. Verify that it can operate on multiple browsers and platforms. Consider using a mobile emulator tool, such as Chrome's [mobile simulator testing tool](#).

Note: *Your team can use additional photographs/images on the web site but ensure that they are royalty free. You can also or alternatively make use of additional AI-generated images.*

The final site and all of its components should be hosted either 1) locally (localhost) or 2) on an accessible staging server. In a submitted screenshare demonstration, you will demonstrate all functionality of your Web application.

Tips & Resources

- [AI Code Generation Tools](#)
- [Creating a React project in Windows](#)
- [CSS Syntax reference](#)
- [CSS Grid and Flexbox](#)
- [Flask and React](#)
- [Responsive Design](#)
- [Responsive Design Best Practices](#)
- [Python and PostgreSQL](#)

Presentation Requirements

- The presentation should be submitted via a recording where you are present and visible on-screen AND your presentation is recorded on your screen. You can find more information on how to record your video presentation [here](#).
- Individuals must submit their presentation recording.
- Groups must also submit a single presentation for their group. Be sure that all group members enable video cameras so that they are visibly present during the recording. All members must speak at least once. This will allow Quantic to verify each participant's identity.
- The demo presentation should last around 5-10 minutes and demonstrate all site functionality, including:
 - Each of the five (or more) pages and navigation between them
 - The email signup for the newsletter
 - A correctly functioning reservations system

- The correct effects of reservations and newsletter signup on the state of the backend database. The effect on the database itself must be shown - not via a site admin page.
 - Discussion of the implementation decisions you have made
- In your presentation, you or each group member must:
 - Present a government-issued ID to the camera so your name and picture are clearly visible and legible.
 - State your name.
- **Please do not “Invite People” in order to share your presentation—there are multiple graders who could potentially be grading your project, and they do not have access to the projects@quantic.edu account.**

Submission Guidelines

Each student or group must submit a single PDF document. For a group, please ensure only ONE member submits on behalf of the group.

The PDF must contain the following links:

- 1) a link to your recorded demo presentation, and
- 2) GitHub repository link(s):
 - a) For an individual, a link to your GitHub repository, or
 - b) For a group, the links to each group member's GitHub repository,

Each GitHub repository must contain the following:

- You **must add** “quantic-grader” as a collaborator to your private repo. Add this collaborator using the repo's Settings > Collaborators > Add people.
- Your private GitHub repository must contain all source code used for your site (frontend and backend). Must include a README.md file in the repo that describes your solution, its design and how to run it locally.
- Must include an [ai-tooling.md](#) file, that contains a summary of any AI tooling you used in developing your solution and how generally you used it (i.e., what worked well, what didn't)
- *Optional*: a [staging.md](#) file, with a link to the staging version (if you have created this) or indicate that you only run it locally

To submit your project, please click on the "Submit Project" button on your dashboard and follow the steps provided in the Google Form. If you are submitting your project as a group, please ensure only ONE member submits on behalf of the group. If you have chosen to work with a group on the web application, you will also be prompted to upload the final page of your [Group Project Agreement](#), which must be completed and signed by all group members. Please reach out to msaie+projects@quantic.edu if you have any

	• The Web site maintains good appearance and evidences excellent UI and UX design • Appropriate use of Flexbox or Grid approaches has been made to achieve a high quality UX • Where forms are needed to meet user requirements, these have been correctly implemented and are working • A back-end Flask app and database correctly integrated with the React front-end and meeting the requirements for the reservation and newsletter signup system • The demo presentation presents all required elements, including the correct effect of reservations and newsletter signups on the backend database and sophisticated reservations logic • A document outlines what AI code generation tools have been used and how
4	Addresses MOST of the project requirements, including: • At least four pages of the Web site have been built using React and JSX • Most of the fundamental functions requested in the SRS have been implemented • The Web site maintains good appearance and evidences good UI and UX design • Appropriate use of Flexbox or Grid approaches has been made to achieve a high quality UX • Where forms are needed to meet user requirements, these have been correctly implemented and are working • A back-end Flask app and database correctly integrated with the React front-end and meeting the requirements for the reservation and newsletter signup system • The demo presentation presents almost all required elements, including the correct effect of reservations and newsletter signups on the backend database • A document outlines what AI code generation tools have been used and how

3	Addresses SOME of the project requirements, including: • Three or more of the pages of the Web site have been built using React and JSX • Some of the fundamental functions requested in the SRS have been implemented • The Web site maintains good appearance and evidences good UI and UX design • Appropriate use of Flexbox or Grid approaches has been made to achieve a high quality UX • Where forms are needed to meet user requirements, these have been correctly implemented and are working • Where forms are needed to meet user requirements, these have been correctly implemented and are working • A back-end Flask app and database correctly integrated with the React front-end and meeting most of the requirements for the reservation and newsletter signup system • The demo presentation presents most required elements, but does not include the correct effect of reservations and newsletter signups on the backend database • A document outlines what AI code generation tools have been used and how
2	Addresses FEW of the project requirements, including: • Two or more of the minimum five pages of the Web site have been built using React and JSX • Few of the fundamental functions requested in the SRS have been implemented • The Web site maintains average appearance and evidences limited UI and UX design • Appropriate use of Flexbox or Grid approaches has been made to achieve a high quality UX • Where forms are needed to meet user requirements, these have been correctly implemented and are partially working • A back-end Flask app and database correctly integrated with the React front-end and meeting few of the requirements for the reservation and newsletter signup system • The demo presentation presents only a few required elements, and does not include the correct effect of reservations and newsletter signups on the backend database

	• A document outlines what AI code generation tools have been used and how
1	Addresses the project but MOST of the project requirements are missing, including: • Less than two of the pages of the Web site have been correctly built using React and JSX • Most of the fundamental functions requested in the SRS have not been implemented • Poor UI and UX design is evidenced • Appropriate use of Box Flexbox or Grid approaches has not been made to achieve a high quality UX • Where forms are needed to meet user requirements, these have not been correctly implemented • A back-end Flask app and database correctly integrated with the React front-end has not been achieved • A very limited or missing demo presentation • A document outlining what AI code generation tools have been used and how is not included
0	• The student either did not complete the assignment, plagiarized all or part of the assignment, or completely failed to address the project requirements.